Plan Effectiveness

CM1010 Introduction to Programming II — Final Submission

How Effective Was My Plan?

178 words

My midterm Gantt chart allocated roughly one week per feature without considering complexity. In practice, the Heart Rate vs Pace scatter plot took twice as long as expected because implementing Pearson's correlation and linear regression required careful mathematical research. Meanwhile, reviewer feedback identified code quality issues — tooltip coupling, missing `let` declarations, and DRY violations — which I had not anticipated in my original timeline.

I revised my plan in three ways. First, I replaced two planned visualisations (Pace Zones and Cumulative Distance) with more distinctive alternatives: a Training Heatmap and Marathon Readiness Radar. Pace Zones overlapped too closely with the existing Pace Progress chart, and the Radar provides a more original multi-dimensional view. Second, I added a dedicated code refactoring phase to address reviewer feedback before building new features. Third, I assigned complexity ratings to each remaining task: the Heatmap is medium complexity (nested date loops), while the Radar is high complexity (trigonometry + normalisation), and allocated time proportionally.

The updated Gantt chart reflects these changes, with buffer time included before the deadline. This revised approach prioritises code quality and originality over feature quantity.

> *See the updated Gantt chart on the following page for the revised project timeline with complexity ratings.*

Updated Gantt Chart — Full Project Timeline

Running Data Visualisation App | Dec 2025 – Mar 2026 | Revised post-midterm with complexity ratings and plan changes

Task	Complexity	Dec W2	Dec W3	Dec W4	Jan W1	Jan W2	Jan W3	Jan W4	Feb W1	Feb W2	Feb W3	Feb W4	Mar W1	Mar W2 (Deadline)
MIDTERM PHASE (Weeks 1–2)														
Planning & Research	Medium													
Core 4 visualisations (BarChart, ScatterPlot, PaceProgress, ActivityTypes)	High													
Midterm submission	Low													
▲ Midterm submitted (12 Dec) — Below: revised final plan ▼														
ORIGINAL PLAN (changed after midterm feedback)														
~~Pace Zones (Distribution Chart)~~	–													
~~Cumulative Distance (Area Chart)~~	–													
REVISED PLAN — New Visualisations														
Goal Tracker (Progress Gauge)	Medium													
Training Heatmap (calendar grid) — *replaces Pace Zones*	Medium													
Marathon Readiness Radar (spider chart) — *replaces Cumulative Distance*	High													
REVISED PLAN — Code Quality (added after reviewer feedback)														
Extract Tooltip to reusable class	Medium													
Fix missing let declarations	Low													
Consolidate duplicate utilities (DRY)	Low													
REVISED PLAN — Documentation & Reports														
Draft Q7, Q2, Q3 reports	Medium													
REVISED PLAN — Testing & Final Submission														
Unit tests + System test cases	Medium													
Finalise Q1, Q4, project logs	Low													
Code PDF + final review + ZIP	Low													

Planning | Midterm Implementation | Documentation | Code Quality | New Feature | Testing & Docs | Final | ~~Cancelled~~ | Low | Medium | High

Key changes from midterm plan:

- **Pace Zones → Training Heatmap:** Pace Zones overlapped with Pace Progress; replaced with a more original GitHub-style calendar visualisation.
- **Cumulative Distance → Marathon Readiness Radar:** Radar provides a multi-dimensional assessment not available in existing tools.
- **Code Quality phase added:** Reviewer feedback identified tooltip coupling, missing `let` declarations, and DRY violations — not anticipated in the original plan.
- **Complexity ratings added:** Midterm plan allocated ~1 week per feature regardless of difficulty. The scatter plot's statistics took 2× longer than expected.